Parul® University
Vadodara, Gujarat

NAAC GRADE A++

Information and Communication Technology

Strings, Packages, and Interfaces

Study Guide

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7.1.1 String Handling

🔷 What is a String?

- A **String** in Java is a sequence of characters.
- Strings are objects of the String class in **java.lang** package.
- They are **immutable**, meaning once created, their value cannot be changed.

🔷 Creating Strings

1. Using String Literals (Stored in String Constant Pool):

```
String str1 = "Hello";
```

2. Using the new Keyword (Stored in Heap memory):

```
String str2 = new String("Hello");
```

🔷 Common String Methods

Method	**Description**	**Example**
length()	Returns length of the string	"Java".length() → 4
charAt(int i)	Returns char at index i	"Java".charAt(2) → 'v'
substring()	Extracts a substring	"Java".substring(1, 3) → "av"
equals()	Compares content (case-sensitive)	"abc".equals("ABC") → false
equalsIgnoreCase()	Compares content ignoring case	"abc".equalsIgnoreCase("ABC") → true
concat()	Appends one string to another	"Hello".concat("World") → "HelloWorld"
compareTo()	Lexicographical comparison	"abc".compareTo("abd") → -1
toLowerCase()	Converts to lowercase	"JAVA".toLowerCase() → "java"
toUpperCase()	Converts to uppercase	"java".toUpperCase() → "JAVA"
trim()	Removes leading and trailing spaces	" Java ".trim() → "Java"

🔷 String Immutability

- Once a String object is created, it **cannot be changed**.
- Any change creates a **new String object**.

```
String s = "Java";
s.concat(" Programming"); // Original 's' still holds "Java"
```

🔷 Memory Management in Strings

- **String Constant Pool:** Stores string literals for reusability and memory efficiency.
- Two identical string literals refer to the same object in the pool.

```
String a = "Java";
String b = "Java";
System.out.println(a == b);  // true
```

🔷 String Comparison

Method	Compares...	Case-sensitive?	Returns
==	References	Yes	true/false
equals()	Values	Yes	true/false
compareTo()	Values lexicographically	Yes	Integer value

7.1.2 Java Packages and Importing

🔷 What is a Package in Java?

- A **package** is a namespace that organizes classes and interfaces in a logical manner.
- Helps **prevent name conflicts** and controls **access**.
- Java packages are similar to folders on your computer.

🔷 Types of Packages

1. **Built-in Packages**
   - Provided by Java API (e.g., java.util, java.io, java.lang)
2. **User-defined Packages**
   - Created by the programmer to group related classes

🔷 Creating a Package

```
// File: MyClass.java
package mypack;

public class MyClass {
   public void display() {
      System.out.println("This is a user-defined package.");
   }
}
```

Using a Package (Importing)

1. **Import a specific class**

```
import mypack.MyClass;
```

2. **Import all classes from a package**

```
import mypack.*;
```

3. **Fully Qualified Name (without import)**

```
mypack.MyClass obj = new mypack.MyClass();
```

The import Statement

- Allows access to classes in other packages
- Must appear at the top of the Java file (after package statement if any)

Setting and Using the Classpath

- **Classpath** is a parameter that tells the JVM and Java compiler where to find class files.
- Set via command line:

```
javac -classpath path MyProgram.java
java -classpath path MyProgram
```

- Can also be set using the environment variable:

```
set CLASSPATH=C:\MyJava\packages
```

Access Modifiers and Packages

Modifier	Same Class	Same Package	Subclass (Different Package)	Other Packages
private	✅	✕	✕	✕
default	✅	✅	✕	✕
protected	✅	✅	✅	✕
public	✅	✅	✅	✅

- **Default (no modifier)** means package-private access.

➢ Summary of Key Concepts:

- Strings in Java are immutable and managed through the String class.
- Java provides a wide range of methods for string manipulation.
- Efficient memory usage is achieved via the String Constant Pool.
- String comparison can be done using equals(), ==, or compareTo().
- Packages help organize and encapsulate classes.
- The `import` keyword is used to include external classes.
- Classpath defines the location of classes to be compiled or executed.
- Access modifiers determine visibility across packages.

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